WEB DEVELOPMENT
ACTIVITY-BASED LEARNING

Three Case Studies • Questions • Solutions • HTML/CSS Code • Output Screenshots

Selected Case Studies

1. Personal Portfolio Website

2. College Information and Admissions Website

3. Online Learning Platform Interface

Submitted by: _____

Roll Number: _____

Course / Section: _____

Submitted to: _____

Introduction

This assignment contains three sample webpages based on the case studies in the supplied Web Development Activity-Based Learning document. Each case study includes the question, solution, rendered output screenshot, and complete HTML with embedded CSS. Save each HTML file and open it in a web browser to run it.

Note: Names, email addresses, course progress, and some college contact details are demonstration content. Replace them with accurate information before publishing. These are front-end examples; backend authentication, real enrollment, and data storage are not implemented.

1. Personal Portfolio Website

Question

Create a personal website displaying a profile, educational qualifications, technical skills, hobbies, projects, achievements, resume, and contact information. Use CSS animations, responsive layouts, and interactive navigation effects.

Solution

A responsive portfolio was created using semantic HTML and CSS. It includes navigation, an introduction, education, skills, projects, hobbies, achievements, and contact information. CSS hover effects and a mobile-friendly layout improve usability.

Output

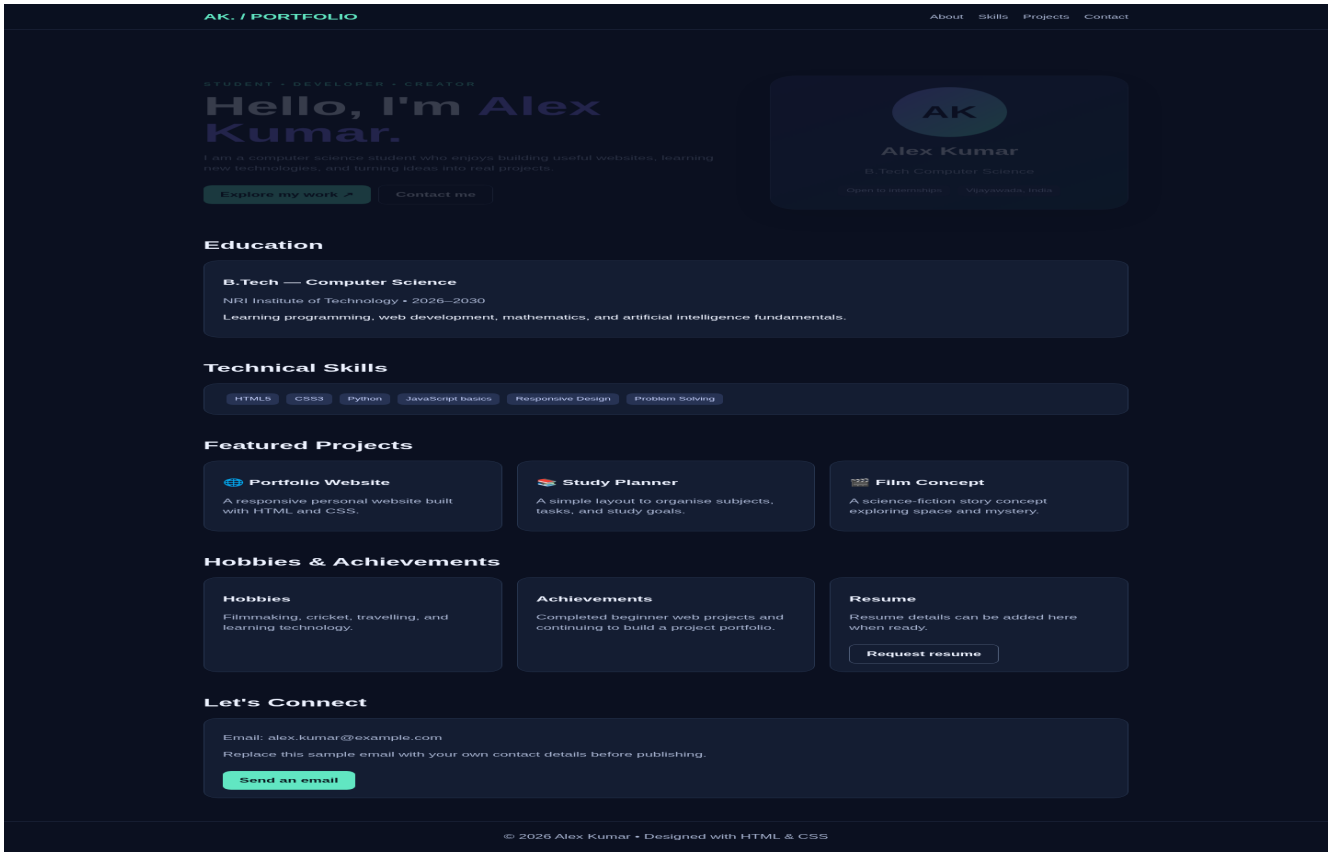


Figure: Rendered webpage screenshot.

1. Personal Portfolio Website — Complete HTML/CSS Code

Copy this code into a file named **01_personal_portfolio.html** and open it in a browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Personal Portfolio | Alex Kumar</title>
<style>
:root{--bg:#0b1020;--panel:#141d32;--text:#edf2ff;--muted:#aab6d3;--accent:#8b7cff;--mint:#61e6c2}
*{box-sizing:border-box;scroll-behavior:smooth}
body{margin:0;font-family:Arial,sans-serif;background:var(--bg);color:var(--text);line-height:1.6}
header{position:sticky;top:0;background:#0b1020ed;border-bottom:1px solid #28334c;z-index:5}
nav{max-width:1050px;margin:auto;padding:16px 22px;display:flex;justify-content:space-between;align-items:
center;gap:18px;flex-wrap:wrap}
.logo{font-size:20px;font-weight:800;color:var(--mint)}nav a{color:var(--muted);text-decoration:none;margin-
left:16px;font-size:14px}
nav a:hover{color:var(--mint)}
main{max-width:1050px;margin:auto;padding:30px 22px}
.hero{padding:70px 0 55px;display:grid;grid-template-columns:1.5fr 1fr;align-items:center;gap:32px;animation:
rise .8s ease-out}
@keyframes rise{from{opacity:0;transform:translateY(18px)}to{opacity:1;transform:translateY(0)}}
.eyebrow{color:var(--mint);text-transform:uppercase;letter-spacing:3px;font-size:12px;font-weight:bold}
h1{font-size:clamp(38px,6vw,64px);line-height:1.05;margin:12px 0}
h1 span{color:var(--accent)}.muted{color:var(--muted)}
.button{display:inline-block;background:var(--mint);color:#08131b;padding:11px 18px;border-radius:9px;text-
decoration:none;font-weight:bold;margin:12px 8px 0 0}
.button.secondary{background:transparent;border:1px solid #53617d;color:var(--text)}
.profile{background:linear-gradient(145deg,#292957,#153b4b);border:1px solid #3a486a;border-radius:28px;
padding:28px;text-align:center;box-shadow:0 20px 50px #0005}
.avatar{width:125px;height:125px;border-radius:50%;display:grid;place-items:center;margin:0 auto 15px;
background:linear-gradient(135deg,var(--accent),var(--mint));color:#10142a;font-size:42px;font-weight:900}
section{padding:28px 0;scroll-margin-top:90px}h2{font-size:27px;margin:0 0 16px}.grid{display:grid;grid-
template-columns:repeat(3,1fr);gap:16px}
.card{background:var(--panel);padding:20px;border:1px solid #283650;border-radius:16px;transition:transform .
2s,border-color .2s}
.card:hover{transform:translateY(-5px);border-color:var(--mint)}.tag{display:inline-block;background:#273354;
color:#c9d4ff;padding:5px 9px;border-radius:7px;margin:4px;font-size:12px}
footer{text-align:center;padding:25px;color:var(--muted);border-top:1px solid #28334c}
@media(max-width:700px){.hero{grid-template-columns:1fr;padding-top:38px}.grid{grid-template-columns:1fr}nav{
justify-content:center}nav a{margin:0 7px}}
</style>
</head>
<body>
<header><nav><div class="logo">AK. / PORTFOLIO</div><div><a href="#about">About</a><a href="#skills">Skills</
a><a href="#projects">Projects</a><a href="#contact">Contact</a></div></nav></header>
<main>
<div class="hero" id="about">
<div><div class="eyebrow">Student • Developer • Creator</div><h1>Hello, I'm <span>Alex Kumar.</span></h1><p
class="muted">I am a computer science student who enjoys building useful websites, learning new technologies,
and turning ideas into real projects.</p><a class="button" href="#projects">Explore my work ■</a><a
class="button secondary" href="#contact">Contact me</a></div>
<div class="profile"><div class="avatar">AK</div><h2 style="margin-bottom:4px">Alex Kumar</h2><p
class="muted">B.Tech Computer Science</p><span class="tag">Open to internships</span><span
class="tag">Vijayawada, India</span></div>
</div>
<section><h2>Education</h2><div class="card"><h3>B.Tech — Computer Science</h3><p class="muted">NRI Institute
of Technology • 2026–2030</p><p>Learning programming, web development, mathematics, and artificial
intelligence fundamentals.</p></div></section>
<section id="skills"><h2>Technical Skills</h2><div class="card"><span class="tag">HTML5</span><span
class="tag">CSS3</span><span class="tag">Python</span><span class="tag">JavaScript basics</span><span
class="tag">Responsive Design</span><span class="tag">Problem Solving</span></div></section>
<section id="projects"><h2>Featured Projects</h2><div class="grid"><article class="card"><h3>■ Portfolio
Website</h3><p class="muted">A responsive personal website built with HTML and CSS.</p></article><article
class="card"><h3>■ Study Planner</h3><p class="muted">A simple layout to organise subjects, tasks, and study
goals.</p></article><article class="card"><h3>■ Film Concept</h3><p class="muted">A science-fiction story
concept exploring space and mystery.</p></article></div></section>
<section><h2>Hobbies & Achievements</h2><div class="grid"><div class="card"><h3>Hobbies</h3><p
class="muted">Filmmaking, cricket, travelling, and learning technology.</p></div><div
class="card"><h3>Achievements</h3><p class="muted">Completed beginner web projects and continuing to build a
project portfolio.</p></div><div class="card"><h3>Resume</h3><p class="muted">Resume details can be added
here when ready.</p><a class="button secondary" href="#contact">Request resume</a></div></div></section>
<section id="contact"><h2>Let's Connect</h2><div class="card"><p class="muted">Email: alex.kumar@example.com</
p><p class="muted">Replace this sample email with your own contact details before publishing.</p><a
class="button" href="mailto:alex.kumar@example.com">Send an email</a></div></section>
</main><footer>© 2026 Alex Kumar • Designed with HTML & CSS</footer>
</body></html>
```

2. College Information and Admissions Website

Question

Design a college website displaying institutional information, departments, courses, faculty profiles, admissions procedures, campus facilities, events, and contact details. Create a navigation bar, banner, and responsive information cards.

Solution

A college information webpage presents an introduction, departments, courses, campus facilities, an admissions process, and contact details. It uses a navigation bar, banner, responsive cards, CSS Grid, and Flexbox.

Output

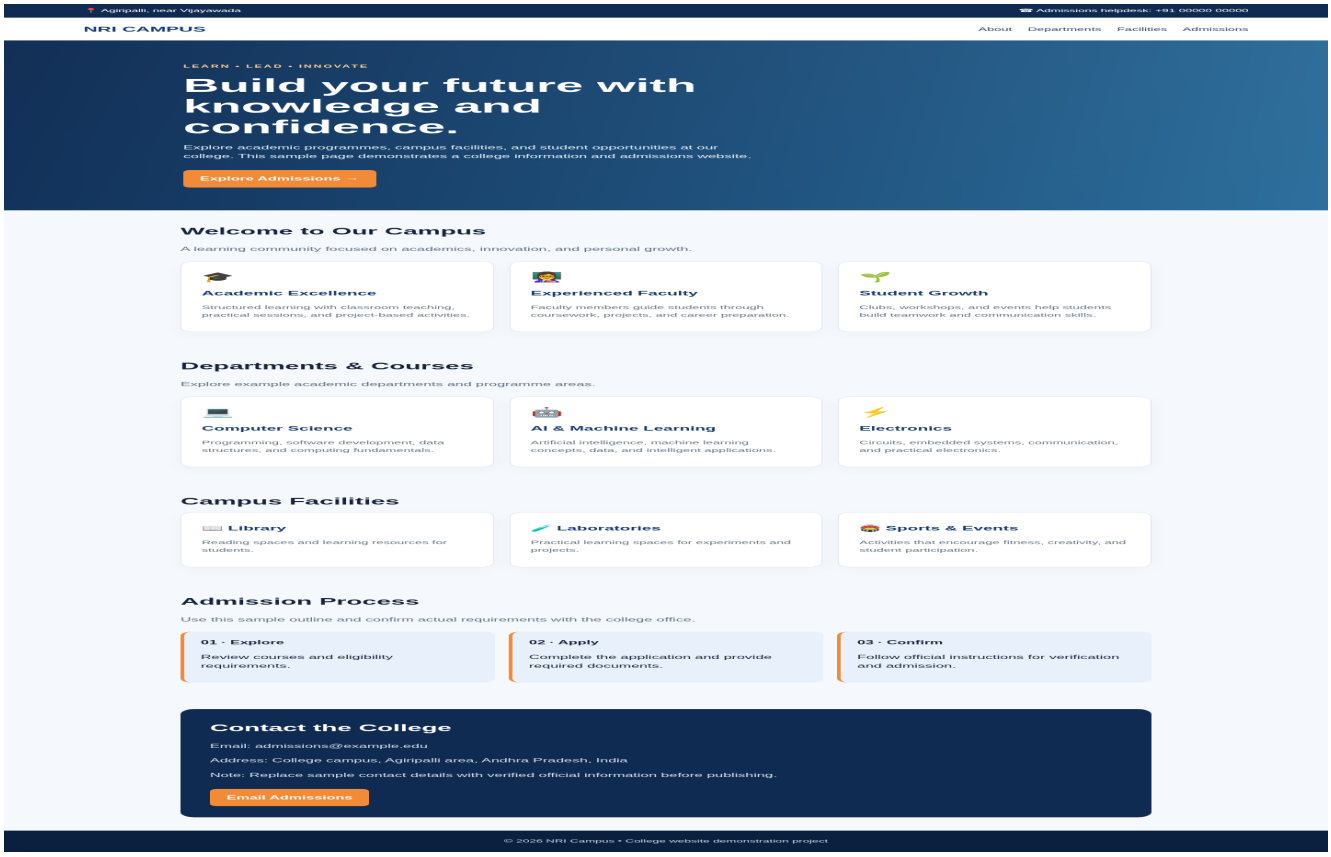


Figure: Rendered webpage screenshot.

2. College Information and Admissions Website — Complete HTML/CSS Code

Copy this code into a file named **02_college_admissions.html** and open it in a browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>NRI Campus | College Information</title>
<style>
*{box-sizing:border-box;scroll-behavior:smooth}body{margin:0;font-family:Arial,sans-serif;color:#17233d;line-
height:1.6;background:#f5f8fc}
.top{background:#102b52;color:white;padding:9px 6%;font-size:13px;display:flex;justify-content:space-between;
gap:10px;flex-wrap:wrap}
nav{padding:16px 6%;display:flex;justify-content:space-between;align-items:center;gap:16px;background:white;
box-shadow:0 3px 15px #17355a10;flex-wrap:wrap}
.brand{font-weight:900;font-size:21px;color:#123c75}nav a{color:#344766;text-decoration:none;margin-left:17px;
font-size:14px}nav a:hover{color:#e47d31}
.hero{padding:65px 6%;color:white;background:linear-gradient(110deg,#102b52ee,#1e6596cc),linear-gradient(
45deg,#224f78,#89b5d4);min-height:320px;display:flex;align-items:center}
.hero-inner{max-width:1050px;margin:auto;width:100%}.label{color:#ffd09a;text-transform:uppercase;letter-
spacing:2px;font-size:12px;font-weight:bold}
h1{font-size:clamp(34px,5vw,54px);line-height:1.1;max-width:680px;margin:14px 0} .hero p{max-width:620px;
color:#e1ebf8}
.btn{display:inline-block;background:#f28b38;color:#fff;text-decoration:none;padding:12px 18px;border-radius:
7px;font-weight:bold;margin-top:10px}
.wrap{max-width:1100px;margin:auto;padding:38px 22px}h2{font-size:28px;margin:0 0 8px}.intro{color:#64738a;
margin-bottom:22px}
.cards{display:grid;grid-template-columns:repeat(3,1fr);gap:17px}.card{background:white;padding:22px;border-
radius:13px;border:1px solid #e0e8f2;box-shadow:0 8px 24px #17355a08;transition:transform .2s}.card:hover{
transform:translateY(-4px)}
.icon{font-size:28px}.card h3{margin:8px 0;color:#143b70}.card p{color:#68768b;font-size:14px}
.steps{display:grid;grid-template-columns:repeat(3,1fr);gap:15px}.step{background:#e8f1fb;padding:18px;border-
radius:12px;border-left:4px solid #f28b38}
.contact{background:#102b52;color:white;padding:32px;border-radius:15px}.contact p{color:#d3dfef}
footer{text-align:center;background:#0b203e;color:#d2deed;padding:20px;font-size:13px}
@media(max-width:720px){.cards,.steps{grid-template-columns:1fr}nav{justify-content:center}nav a{margin:0
7px}.top{justify-content:center}.hero{padding:45px 7%}}
</style>
</head>
<body>
<div class="top"><span>■ Agiripalli, near Vijayawada</span><span>✎ Admissions helpdesk: +91 00000 00000</
span></div>
<nav><div class="brand">NRI CAMPUS</div><div><a href="#about">About</a><a href="#departments">Departments</
a><a href="#facilities">Facilities</a><a href="#admissions">Admissions</a></div></nav>
<section class="hero"><div class="hero-inner"><div class="label">Learn • Lead • Innovate</div><h1>Build your
future with knowledge and confidence.</h1><p>Explore academic programmes, campus facilities, and student
opportunities at our college. This sample page demonstrates a college information and admissions website.</
p><a class="btn" href="#admissions">Explore Admissions →</a></div></section>
<main class="wrap" id="about"><h2>Welcome to Our Campus</h2><p class="intro">A learning community focused on
academics, innovation, and personal growth.</p><div class="cards"><article class="card"><div class="icon">■</
div><h3>Academic Excellence</h3><p>Structured learning with classroom teaching, practical sessions, and
project-based activities.</p></article><article class="card"><div class="icon">■■■</div><h3>Experienced
Faculty</h3><p>Faculty members guide students through coursework, projects, and career preparation.</p></
article><article class="card"><div class="icon">■</div><h3>Student Growth</h3><p>Clubs, workshops, and events
help students build teamwork and communication skills.</p></article></div></main>
<section class="wrap" id="departments"><h2>Departments & Courses</h2><p class="intro">Explore example
academic departments and programme areas.</p><div class="cards"><article class="card"><div class="icon">■</
div><h3>Computer Science</h3><p>Programming, software development, data structures, and computing
fundamentals.</p></article><article class="card"><div class="icon">■</div><h3>AI & Machine Learning</
h3><p>Artificial intelligence, machine learning concepts, data, and intelligent applications.</p></
article><article class="card"><div class="icon">■</div><h3>Electronics</h3><p>Circuits, embedded systems,
communication, and practical electronics.</p></article></div></section>
<section class="wrap" id="facilities"><h2>Campus Facilities</h2><div class="cards"><article
class="card"><h3>■ Library</h3><p>Reading spaces and learning resources for students.</p></article><article
class="card"><h3>■ Laboratories</h3><p>Practical learning spaces for experiments and projects.</p></
article><article class="card"><h3>■■ Sports & Events</h3><p>Activities that encourage fitness, creativity,
and student participation.</p></article></div></section>
<section class="wrap" id="admissions"><h2>Admission Process</h2><p class="intro">Use this sample outline and
confirm actual requirements with the college office.</p><div class="steps"><div class="step"><b>01 · Explore</
b><p>Review courses and eligibility requirements.</p></div><div class="step"><b>02 · Apply</b><p>Complete the
application and provide required documents.</p></div><div class="step"><b>03 · Confirm</b><p>Follow official
instructions for verification and admission.</p></div></div></section>
<section class="wrap"><div class="contact"><h2>Contact the College</h2><p>Email: admissions@example.edu</
p><p>Address: College campus, Agiripalli area, Andhra Pradesh, India</p><p>Note: Replace sample contact
details with verified official information before publishing.</p><a class="btn" href="mailto:
admissions@example.edu">Email Admissions</a></div></section>
<footer>© 2026 NRI Campus • College website demonstration project</footer>
</body></html>
```

3. Online Learning Platform Interface

Question

Develop an e-learning website containing course categories, video lesson thumbnails, instructor profiles, course descriptions, learning objectives, and enrollment buttons. Design responsive course cards and progress indicators.

Solution

An online learning interface includes course categories, course cards, instructor names, progress bars, and course buttons. Responsive CSS Grid adapts the catalogue to different screen sizes. Real enrollment and progress storage would require additional JavaScript and/or a backend.

Output

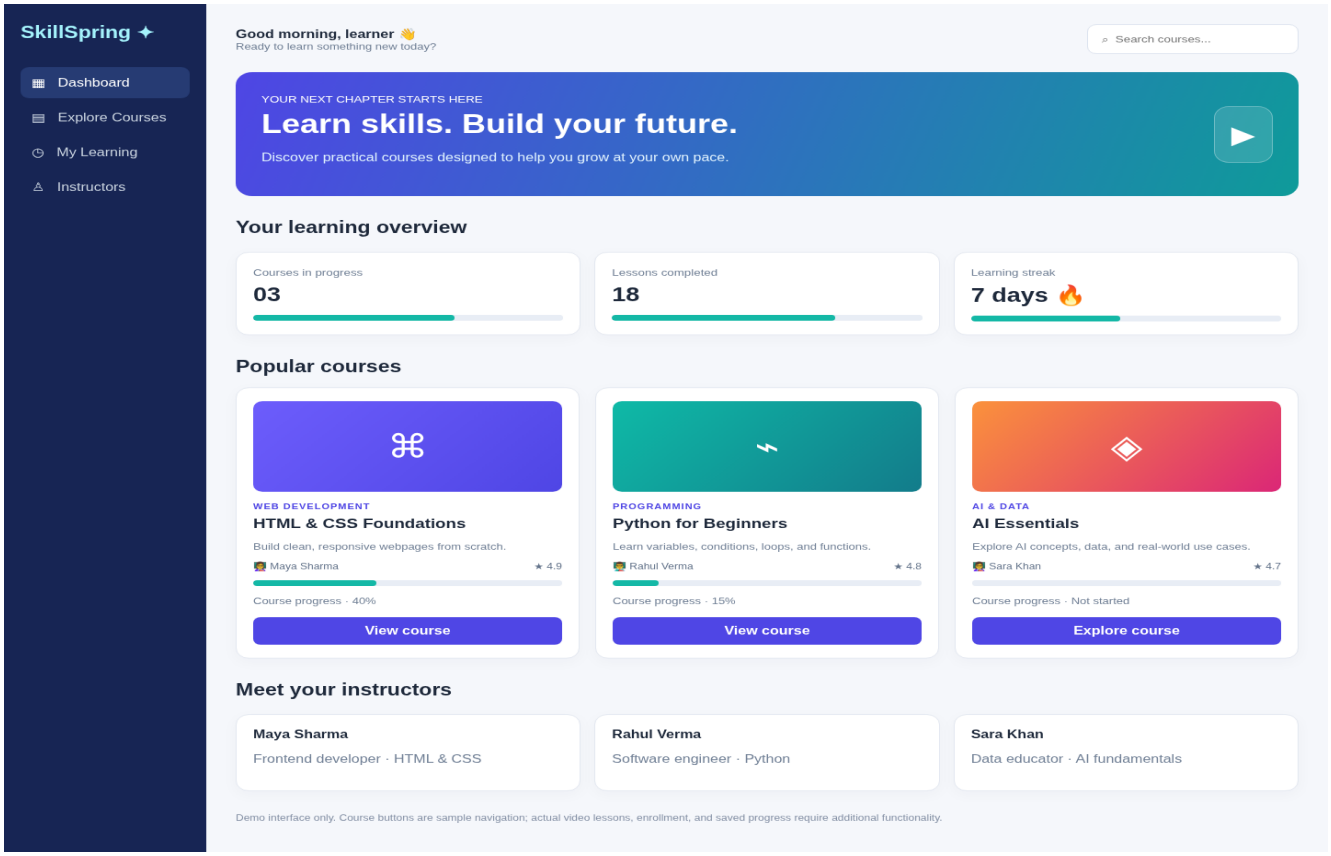


Figure: Rendered webpage screenshot.

3. Online Learning Platform Interface — Complete HTML/CSS Code

Copy this code into a file named **03_online_learning.html** and open it in a browser.

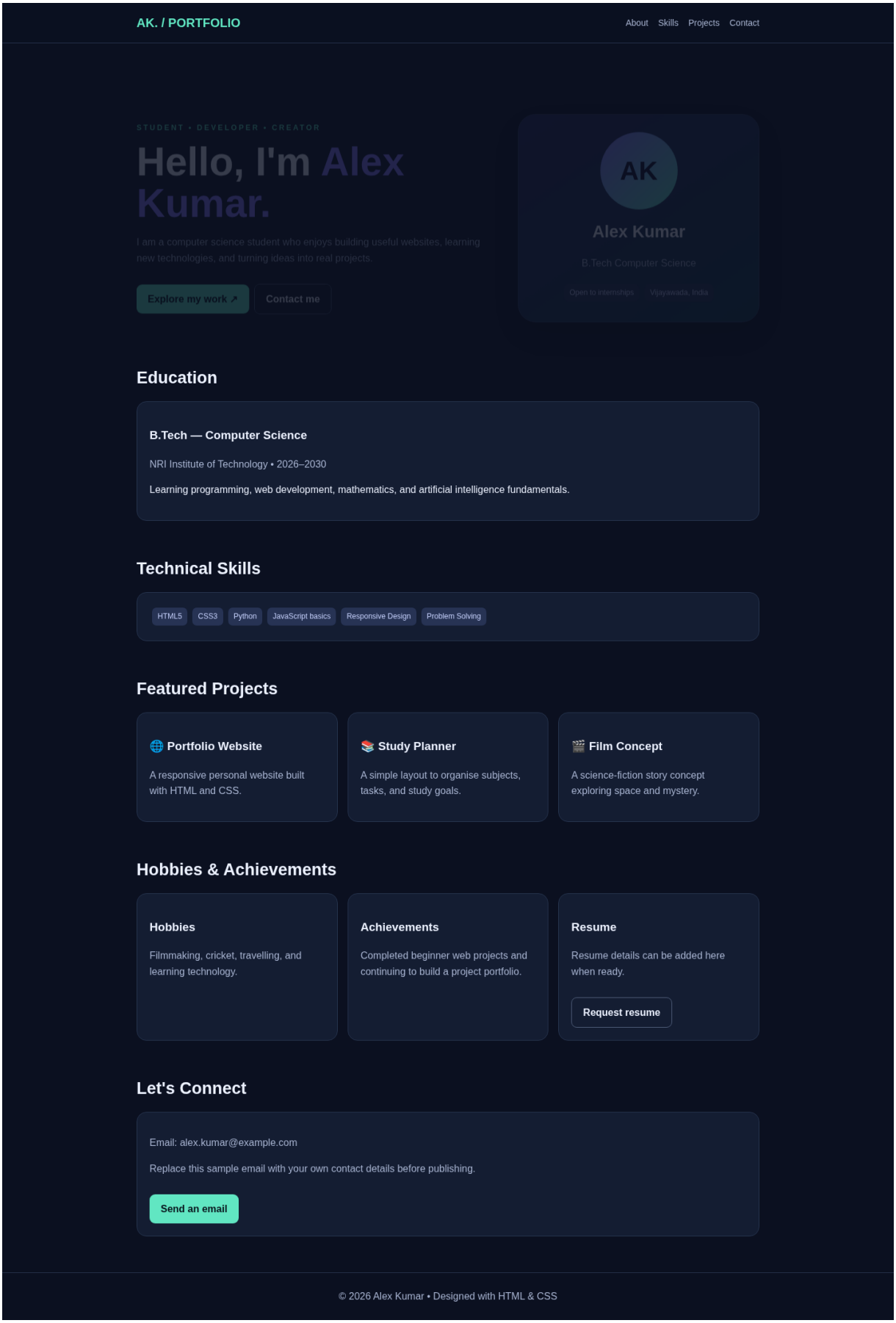
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SkillSpring | Online Learning</title>
<style>
*{box-sizing:border-box}body{margin:0;font-family:Arial,sans-serif;background:#f5f7fb;color:#1e293b}
.layout{display:grid;grid-template-columns:220px 1fr;min-height:100vh}.sidebar{background:#172554;color:
#dbeafe;padding:26px 18px}.brand{font-size:23px;font-weight:900;color:#a5f3fc;margin-bottom:34px}.side-link{
display:block;text-decoration:none;color:#cbd5e1;padding:12px;border-radius:9px;margin:5px 0}.side-link.
active,.side-link:hover{background:#263b73;color:white}
main{padding:28px 32px;max-width:1300px;width:100%;margin:auto}.topbar{display:flex;justify-content:space-
between;align-items:center;gap:15px;flex-wrap:wrap;margin-bottom:25px}.search{background:white;border:1px
solid #dce3ef;border-radius:10px;padding:12px 15px;min-width:230px}
.hero{background:linear-gradient(110deg,#4f46e5,#0f9b9a);border-radius:18px;padding:28px;color:white;display:
flex;justify-content:space-between;align-items:center;gap:20px}.hero h1{font-size:clamp(25px,4vw,36px);margin:
5px 0}.hero p{color:#e0f2fe;max-width:570px}.hero-badge{background:#ffffff22;border:1px solid #ffffff50;
border-radius:16px;padding:18px;font-size:34px}
h2{font-size:23px;margin:30px 0 15px}.stats{display:grid;grid-template-columns:repeat(3,1fr);gap:15px;margin-
top:20px}.stat,.course{background:white;border:1px solid #e3e8f1;border-radius:14px;padding:18px;box-shadow:0
5px 18px #1d2b4a08}.stat small{color:#718096}.number{font-size:27px;font-weight:800;margin-top:7px}
.progress{height:8px;background:#e8edf5;border-radius:99px;overflow:hidden;margin-top:12px}.progress span{
display:block;height:100%;background:#14b8a6;border-radius:99px}
.courses{display:grid;grid-template-columns:repeat(3,1fr);gap:17px}.thumb{height:125px;border-radius:10px;
display:grid;place-items:center;font-size:45px;color:white;background:linear-gradient(135deg,#6d5dfc,
#4f46e5)}.thumb.green{background:linear-gradient(135deg,#0fbaa7,#137c8b)}.thumb.orange{background:linear-
gradient(135deg,#fb923c,#db2777)}
.category{font-size:11px;font-weight:bold;letter-spacing:1px;color:#4f46e5;margin-top:14px}.course h3{margin:
7px 0}.course p{font-size:13px;color:#718096}.meta{display:flex;justify-content:space-between;gap:8px;font-
size:12px;color:#64748b}.enroll{display:block;text-align:center;text-decoration:none;background:#4f46e5;color:
white;padding:10px;border-radius:8px;margin-top:15px;font-weight:bold}.enroll:hover{background:#3730a3}
@media(max-width:850px){.layout{grid-template-columns:1fr}.sidebar{padding:14px;display:flex;align-items:
center;gap:10px;flex-wrap:wrap}.brand{margin:0 15px 0 0}.side-link{display:inline-block;margin:0;padding:8px}.
courses{grid-template-columns:repeat(2,1fr)}main{padding:22px}}
@media(max-width:560px){.courses,.stats{grid-template-columns:1fr}.hero{align-items:flex-start}.hero-badge{
display:none}.search{width:100%}}
</style>
</head>
<body><div class="layout">
<aside class="sidebar"><div class="brand">SkillSpring ✦</div><a class="side-link active" href="#home">■  
 Dashboard</a><a class="side-link" href="#courses">■   Explore Courses</a><a class="side-link"
href="#progress">■   My Learning</a><a class="side-link" href="#instructors">■   Instructors</a></
aside>
<main id="home"><div class="topbar"><div><b>Good morning, learner ■</b><div style="font-size:13px;color:
#718096">Ready to learn something new today?</div></div><input class="search" aria-label="Search courses"
placeholder="■  Search courses..."></div>
<section class="hero"><div><small>YOUR NEXT CHAPTER STARTS HERE</small><h1>Learn skills. Build your future.</
h1><p>Discover practical courses designed to help you grow at your own pace.</p></div><div class="hero-
badge">■</div></section>
<section id="progress"><h2>Your learning overview</h2><div class="stats"><div class="stat"><small>Courses in
progress</small><div class="number">03</div><div class="progress"><span style="width:65%"></span></div></
div><div class="stat"><small>Lessons completed</small><div class="number">18</div><div class="progress"><span
style="width:72%"></span></div></div><div class="stat"><small>Learning streak</small><div class="number">7
days ■</div><div class="progress"><span style="width:48%"></span></div></div></div></section>
<section id="courses"><h2>Popular courses</h2><div class="courses">
<article class="course"><div class="thumb">■</div><div class="category">WEB DEVELOPMENT</div><h3>HTML & CSS
Foundations</h3><p>Build clean, responsive webpages from scratch.</p><div class="meta"><span>■■■ Maya Sharma</
span><span>★ 4.9</span></div><div class="progress"><span style="width:40%"></span></div><p>Course progress ·
40%</p><a class="enroll" href="#instructors">View course</a></article>
<article class="course"><div class="thumb green">■</div><div class="category">PROGRAMMING</div><h3>Python for
Beginners</h3><p>Learn variables, conditions, loops, and functions.</p><div class="meta"><span>■■■ Rahul
Verma</span><span>★ 4.8</span></div><div class="progress"><span style="width:15%"></span></div><p>Course
progress · 15%</p><a class="enroll" href="#instructors">View course</a></article>
<article class="course"><div class="thumb orange">■</div><div class="category">AI & DATA</div><h3>AI
Essentials</h3><p>Explore AI concepts, data, and real-world use cases.</p><div class="meta"><span>■■■ Sara
Khan</span><span>★ 4.7</span></div><div class="progress"><span style="width:0%"></span></div><p>Course
progress · Not started</p><a class="enroll" href="#instructors">Explore course</a></article>
</div></section>
<section id="instructors"><h2>Meet your instructors</h2><div class="stats"><div class="stat"><b>Maya Sharma</
b><p style="color:#718096">Frontend developer · HTML & CSS</p></div><div class="stat"><b>Rahul Verma</b><p
style="color:#718096">Software engineer · Python</p></div><div class="stat"><b>Sara Khan</b><p style="color:
#718096">Data educator · AI fundamentals</p></div></div></section>
<p style="font-size:12px;color:#8490a5;margin-top:30px">Demo interface only. Course buttons are sample
navigation; actual video lessons, enrollment, and saved progress require additional functionality.</p>
</main></div></body></html>
```

APPENDIX — WEBPAGE OUTPUTS AND COMPLETE SOURCE CODE

This appendix contains the rendered webpage screenshots and complete HTML/CSS source code for the three selected concepts.

1. Personal Portfolio Website

Webpage output screenshot



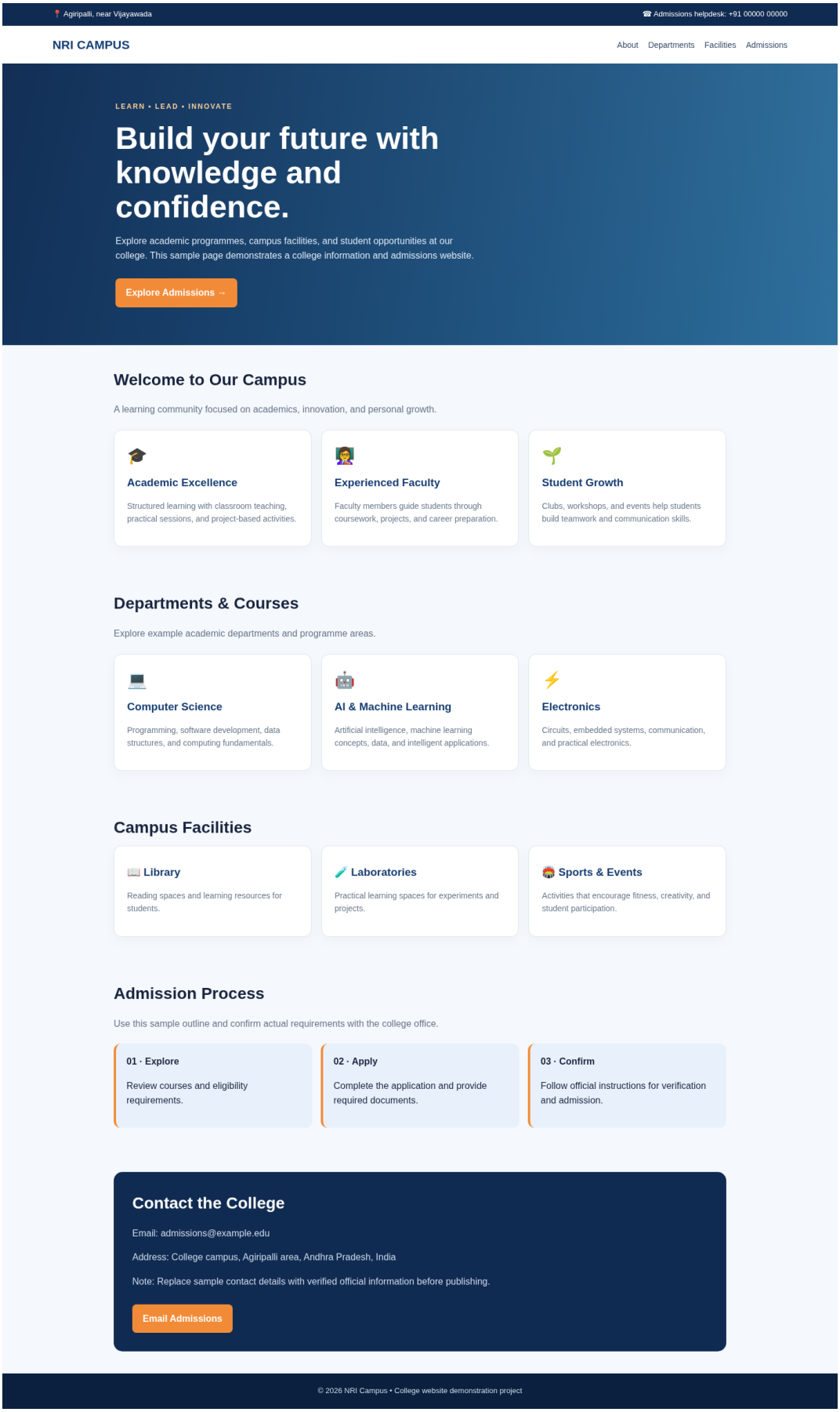
1. Personal Portfolio Website — HTML/CSS code

Save the following code as **01_personal_portfolio.html** and open it in a web browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Personal Portfolio | Alex Kumar</title>
<style>
:root{--bg:#0b1020;--panel:#141d32;--text:#edf2ff;--muted:#aab6d3;--accent:#8b7cff;--mint:#61e6c2}
*{box-sizing:border-box;scroll-behavior:smooth}
body{margin:0;font-family:Arial,sans-serif;background:var(--bg);color:var(--text);line-height:1.6}
header{position:sticky;top:0;background:#0b1020ed;border-bottom:1px solid #28334c;z-index:5}
nav{max-width:1050px;margin:auto;padding:16px 22px;display:flex;justify-content:space-between;align-
items:center;gap:18px;flex-wrap:wrap}
.logo{font-size:20px;font-weight:800;color:var(--mint)}nav a{color:var(--muted);text-decoration:none;
margin-left:16px;font-size:14px}
nav a:hover{color:var(--mint)}
main{max-width:1050px;margin:auto;padding:30px 22px}
.hero{padding:70px 0 55px;display:grid;grid-template-columns:1.5fr 1fr;align-items:center;gap:32px;
animation:rise .8s ease-out}
@keyframes rise{from{opacity:0;transform:translateY(18px)}to{opacity:1;transform:translateY(0)}}
.eyebrow{color:var(--mint);text-transform:uppercase;letter-spacing:3px;font-size:12px;font-weight:bold}
h1{font-size:clamp(38px,6vw,64px);line-height:1.05;margin:12px 0}
h1 span{color:var(--accent)}.muted{color:var(--muted)}
.button{display:inline-block;background:var(--mint);color:#08131b;padding:11px 18px;border-radius:9px;
text-decoration:none;font-weight:bold;margin:12px 8px 0 0}
.button.secondary{background:transparent;border:1px solid #53617d;color:var(--text)}
.profile{background:linear-gradient(145deg,#292957,#153b4b);border:1px solid #3a486a;border-radius:28px;
padding:28px;text-align:center;box-shadow:0 20px 50px #0005}
.avatar{width:125px;height:125px;border-radius:50%;display:grid;place-items:center;margin:0 auto 15px;
background:linear-gradient(135deg,var(--accent),var(--mint));color:#10142a;font-size:42px;font-weight:
900}
section{padding:28px 0;scroll-margin-top:90px}h2{font-size:27px;margin:0 0 16px}.grid{display:grid;grid-
template-columns:repeat(3,1fr);gap:16px}
.card{background:var(--panel);padding:20px;border:1px solid #283650;border-radius:16px;transition:
transform .2s,border-color .2s}
.card:hover{transform:translateY(-5px);border-color:var(--mint)}.tag{display:inline-block;background:
#273354;color:#c9d4ff;padding:5px 9px;border-radius:7px;margin:4px;font-size:12px}
footer{text-align:center;padding:25px;color:var(--muted);border-top:1px solid #28334c}
@media(max-width:700px){.hero{grid-template-columns:1fr;padding-top:38px}.grid{grid-template-columns:
1fr}nav{justify-content:center}nav a{margin:0 7px}}
</style>
</head>
<body>
<header><nav><div class="logo">AK. / PORTFOLIO</div><div><a href="#about">About</a><a
href="#skills">Skills</a><a href="#projects">Projects</a><a href="#contact">Contact</a></div></nav></
header>
<main>
<div class="hero" id="about">
<div><div class="eyebrow">Student • Developer • Creator</div><h1>Hello, I'm <span>Alex Kumar.</span></
h1><p class="muted">I am a computer science student who enjoys building useful websites, learning new
technologies, and turning ideas into real projects.</p><a class="button" href="#projects">Explore my
work ■</a><a class="button secondary" href="#contact">Contact me</a></div>
<div class="profile"><div class="avatar">AK</div><h2 style="margin-bottom:4px">Alex Kumar</h2><p
class="muted">B.Tech Computer Science</p><span class="tag">Open to internships</span><span
class="tag">Vijayawada, India</span></div>
</div>
<section><h2>Education</h2><div class="card"><h3>B.Tech — Computer Science</h3><p class="muted">NRI
Institute of Technology • 2026–2030</p><p>Learning programming, web development, mathematics, and
artificial intelligence fundamentals.</p></div></section>
<section id="skills"><h2>Technical Skills</h2><div class="card"><span class="tag">HTML5</span><span
class="tag">CSS3</span><span class="tag">Python</span><span class="tag">JavaScript basics</span><span
class="tag">Responsive Design</span><span class="tag">Problem Solving</span></div></section>
<section id="projects"><h2>Featured Projects</h2><div class="grid"><article class="card"><h3>■ Portfolio
Website</h3><p class="muted">A responsive personal website built with HTML and CSS.</p></
article><article class="card"><h3>■ Study Planner</h3><p class="muted">A simple layout to organise
subjects, tasks, and study goals.</p></article><article class="card"><h3>■ Film Concept</h3><p
class="muted">A science-fiction story concept exploring space and mystery.</p></article></div></section>
<section><h2>Hobbies & Achievements</h2><div class="grid"><div class="card"><h3>Hobbies</h3><p
class="muted">Filmmaking, cricket, travelling, and learning technology.</p></div><div
class="card"><h3>Achievements</h3><p class="muted">Completed beginner web projects and continuing to
build a project portfolio.</p></div><div class="card"><h3>Resume</h3><p class="muted">Resume details can
be added here when ready.</p><a class="button secondary" href="#contact">Request resume</a></div></div></
section>
<section id="contact"><h2>Let's Connect</h2><div class="card"><p class="muted">Email: alex.kumar@example.
com</p><p class="muted">Replace this sample email with your own contact details before publishing.</p><a
class="button" href="mailto:alex.kumar@example.com">Send an email</a></div></section>
</main><footer>© 2026 Alex Kumar • Designed with HTML & CSS</footer>
</body></html>
```

2. College Information and Admissions Website

Webpage output screenshot



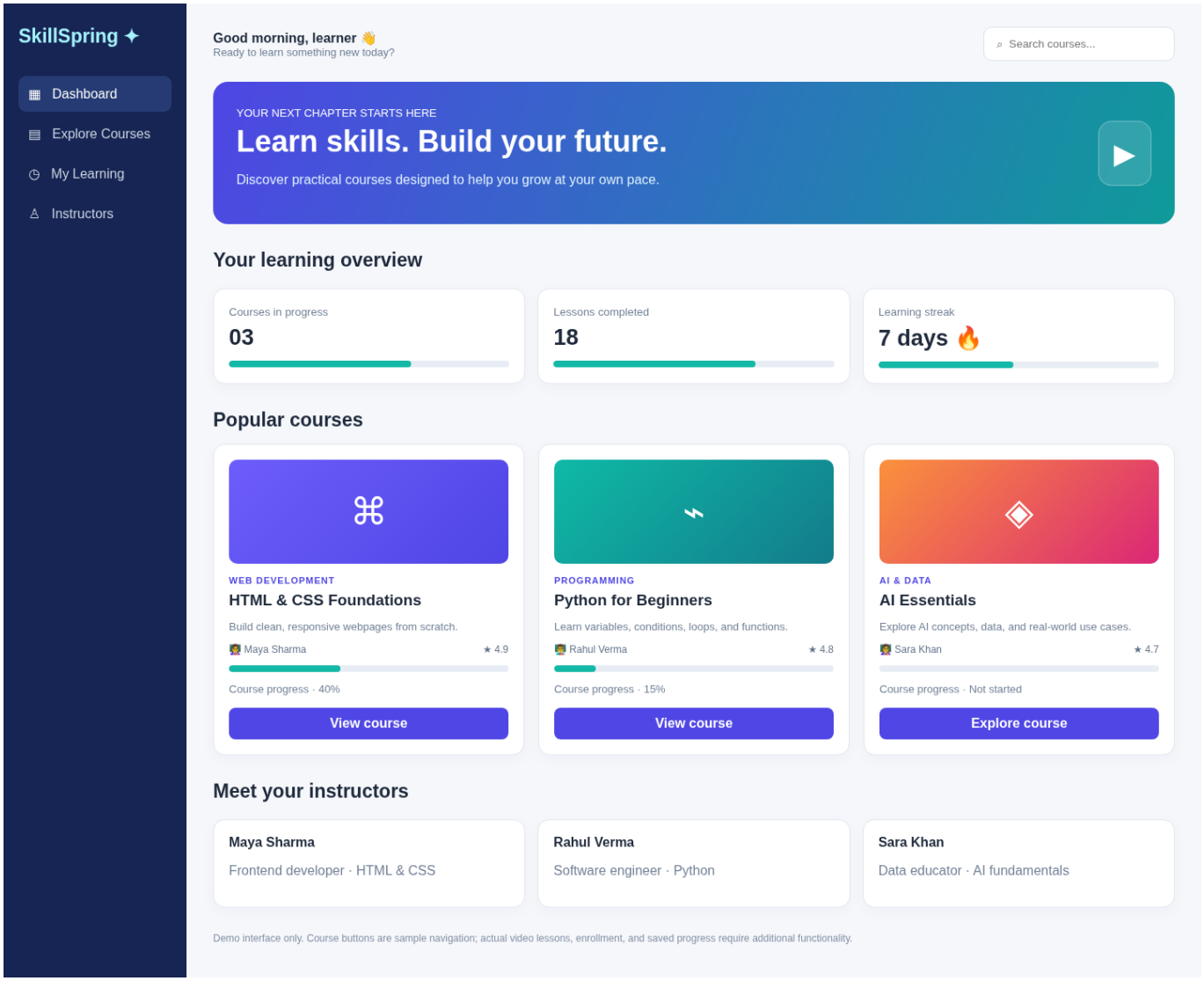
2. College Information and Admissions Website — HTML/CSS code

Save the following code as **02_college_admissions.html** and open it in a web browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>NRI Campus | College Information</title>
<style>
*{box-sizing:border-box;scroll-behavior:smooth}body{margin:0;font-family:Arial,sans-serif;color:#17233d;
line-height:1.6;background:#f5f8fc}
.top{background:#102b52;color:white;padding:9px 6%;font-size:13px;display:flex;justify-content:space-
between;gap:10px;flex-wrap:wrap}
nav{padding:16px 6%;display:flex;justify-content:space-between;align-items:center;gap:16px;background:
white;box-shadow:0 3px 15px #17355a10;flex-wrap:wrap}
.brand{font-weight:900;font-size:21px;color:#123c75}nav a{color:#344766;text-decoration:none;margin-left:
17px;font-size:14px}nav a:hover{color:#e47d31}
.hero{padding:65px 6%;color:white;background:linear-gradient(110deg,#102b52ee,#1e6596cc),linear-gradient(
45deg,#224f78,#89b5d4);min-height:320px;display:flex;align-items:center}
.hero-inner{max-width:1050px;margin:auto;width:100%}.label{color:#ffd09a;text-transform:uppercase;letter-
spacing:2px;font-size:12px;font-weight:bold}
h1{font-size:clamp(34px,5vw,54px);line-height:1.1;max-width:680px;margin:14px 0} .hero p{max-width:620px;
color:#e1ebf8}
.btn{display:inline-block;background:#f28b38;color:#fff;text-decoration:none;padding:12px 18px;border-
radius:7px;font-weight:bold;margin-top:10px}
.wrap{max-width:1100px;margin:auto;padding:38px 22px}h2{font-size:28px;margin:0 0 8px}.intro{color:
#64738a;margin-bottom:22px}
.cards{display:grid;grid-template-columns:repeat(3,1fr);gap:17px}.card{background:white;padding:22px;
border-radius:13px;border:1px solid #e0e8f2;box-shadow:0 8px 24px #17355a08;transition:transform .2s}.
card:hover{transform:translateY(-4px)}
.icon{font-size:28px}.card h3{margin:8px 0;color:#143b70}.card p{color:#68768b;font-size:14px}
.steps{display:grid;grid-template-columns:repeat(3,1fr);gap:15px}.step{background:#e8f1fb;padding:18px;
border-radius:12px;border-left:4px solid #f28b38}
.contact{background:#102b52;color:white;padding:32px;border-radius:15px}.contact p{color:#d3dfef}
footer{text-align:center;background:#0b203e;color:#d2deed;padding:20px;font-size:13px}
@media(max-width:720px){.cards,.steps{grid-template-columns:1fr}nav{justify-content:center}nav a{margin:
0 7px}.top{justify-content:center}.hero{padding:45px 7%}}
</style>
</head>
<body>
<div class="top"><span>■ Agiripalli, near Vijayawada</span><span>✆ Admissions helpdesk: +91 00000 00000</
span></div>
<nav><div class="brand">NRI CAMPUS</div><div><a href="#about">About</a><a
href="#departments">Departments</a><a href="#facilities">Facilities</a><a href="#admissions">Admissions</
a></div></nav>
<section class="hero"><div class="hero-inner"><div class="label">Learn • Lead • Innovate</div><h1>Build
your future with knowledge and confidence.</h1><p>Explore academic programmes, campus facilities, and
student opportunities at our college. This sample page demonstrates a college information and admissions
website.</p><a class="btn" href="#admissions">Explore Admissions →</a></div></section>
<main class="wrap" id="about"><h2>Welcome to Our Campus</h2><p class="intro">A learning community
focused on academics, innovation, and personal growth.</p><div class="cards"><article class="card"><div
class="icon">■</div><h3>Academic Excellence</h3><p>Structured learning with classroom teaching,
practical sessions, and project-based activities.</p></article><article class="card"><div
class="icon">■■■</div><h3>Experienced Faculty</h3><p>Faculty members guide students through coursework,
projects, and career preparation.</p></article><article class="card"><div class="icon">■</
div><h3>Student Growth</h3><p>Clubs, workshops, and events help students build teamwork and
communication skills.</p></article></div></main>
<section class="wrap" id="departments"><h2>Departments & Courses</h2><p class="intro">Explore example
academic departments and programme areas.</p><div class="cards"><article class="card"><div
class="icon">■</div><h3>Computer Science</h3><p>Programming, software development, data structures, and
computing fundamentals.</p></article><article class="card"><div class="icon">■</div><h3>AI & Machine
Learning</h3><p>Artificial intelligence, machine learning concepts, data, and intelligent applications.</
p></article><article class="card"><div class="icon">■</div><h3>Electronics</h3><p>Circuits, embedded
systems, communication, and practical electronics.</p></article></div></section>
<section class="wrap" id="facilities"><h2>Campus Facilities</h2><div class="cards"><article
class="card"><h3>■ Library</h3><p>Reading spaces and learning resources for students.</p></
article><article class="card"><h3>■ Laboratories</h3><p>Practical learning spaces for experiments and
projects.</p></article><article class="card"><h3>■■ Sports & Events</h3><p>Activities that encourage
fitness, creativity, and student participation.</p></article></div></section>
<section class="wrap" id="admissions"><h2>Admission Process</h2><p class="intro">Use this sample outline
and confirm actual requirements with the college office.</p><div class="steps"><div class="step"><b>01 ·
Explore</b><p>Review courses and eligibility requirements.</p></div><div class="step"><b>02 · Apply</
b><p>Complete the application and provide required documents.</p></div><div class="step"><b>03 ·
Confirm</b><p>Follow official instructions for verification and admission.</p></div></div></section>
<section class="wrap"><div class="contact"><h2>Contact the College</h2><p>Email: admissions@example.edu</
p><p>Address: College campus, Agiripalli area, Andhra Pradesh, India</p><p>Note: Replace sample contact
details with verified official information before publishing.</p><a class="btn" href="mailto:
admissions@example.edu">Email Admissions</a></div></section>
<footer>© 2026 NRI Campus • College website demonstration project</footer>
</body></html>
```

3. Online Learning Platform Interface

Webpage output screenshot



3. Online Learning Platform Interface — HTML/CSS code

Save the following code as **03_online_learning.html** and open it in a web browser.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SkillSpring | Online Learning</title>
<style>
*{box-sizing:border-box}body{margin:0;font-family:Arial,sans-serif;background:#f5f7fb;color:#1e293b}
.layout{display:grid;grid-template-columns:220px 1fr;min-height:100vh}.sidebar{background:#172554;color:
#dbeafe;padding:26px 18px}.brand{font-size:23px;font-weight:900;color:#a5f3fc;margin-bottom:34px}.side-
link{display:block;text-decoration:none;color:#cbd5e1;padding:12px;border-radius:9px;margin:5px 0}.side-
link.active,.side-link:hover{background:#263b73;color:white}
main{padding:28px 32px;max-width:1300px;width:100%;margin:auto}.topbar{display:flex;justify-content:
space-between;align-items:center;gap:15px;flex-wrap:wrap;margin-bottom:25px}.search{background:white;
border:1px solid #dce3ef;border-radius:10px;padding:12px 15px;min-width:230px}
.hero{background:linear-gradient(110deg,#4f46e5,#0f9b9a);border-radius:18px;padding:28px;color:white;
display:flex;justify-content:space-between;align-items:center;gap:20px}.hero h1{font-size:clamp(25px,4vw,
36px);margin:5px 0}.hero p{color:#e0f2fe;max-width:570px}.hero-badge{background:#ffffff22;border:1px
solid #ffffff50;border-radius:16px;padding:18px;font-size:34px}
h2{font-size:23px;margin:30px 0 15px}.stats{display:grid;grid-template-columns:repeat(3,1fr);gap:15px;
margin-top:20px}.stat,.course{background:white;border:1px solid #e3e8f1;border-radius:14px;padding:18px;
box-shadow:0 5px 18px #1d2b4a08}.stat small{color:#718096}.number{font-size:27px;font-weight:800;margin-
top:7px}
.progress{height:8px;background:#e8edf5;border-radius:99px;overflow:hidden;margin-top:12px}.progress
span{display:block;height:100%;background:#14b8a6;border-radius:99px}
.courses{display:grid;grid-template-columns:repeat(3,1fr);gap:17px}.thumb{height:125px;border-radius:
10px;display:grid;place-items:center;font-size:45px;color:white;background:linear-gradient(135deg,
#6d5dfc,#4f46e5)}.thumb.green{background:linear-gradient(135deg,#0fbaa7,#137c8b)}.thumb.orange{
background:linear-gradient(135deg,#fb923c,#db2777)}
.category{font-size:11px;font-weight:bold;letter-spacing:1px;color:#4f46e5;margin-top:14px}.course h3{
margin:7px 0}.course p{font-size:13px;color:#718096}.meta{display:flex;justify-content:space-between;gap:
8px;font-size:12px;color:#64748b}.enroll{display:block;text-align:center;text-decoration:none;background:
#4f46e5;color:white;padding:10px;border-radius:8px;margin-top:15px;font-weight:bold}.enroll:hover{
background:#3730a3}
@media(max-width:850px){.layout{grid-template-columns:1fr}.sidebar{padding:14px;display:flex;align-items:
center;gap:10px;flex-wrap:wrap}.brand{margin:0 15px 0 0}.side-link{display:inline-block;margin:0;padding:
8px}.courses{grid-template-columns:repeat(2,1fr)}main{padding:22px}}
@media(max-width:560px){.courses,.stats{grid-template-columns:1fr}.hero{align-items:flex-start}.hero-
badge{display:none}.search{width:100%}}
</style>
</head>
<body><div class="layout">
<aside class="sidebar"><div class="brand">SkillSpring ✦</div><a class="side-link active" href="#home">■
  Dashboard</a><a class="side-link" href="#courses">■   Explore Courses</a><a class="side-
link" href="#progress">■   My Learning</a><a class="side-link" href="#instructors">■  
Instructors</a></aside>
<main id="home"><div class="topbar"><div><b>Good morning, learner ■</b><div style="font-size:13px;color:
#718096">Ready to learn something new today?</div></div><input class="search" aria-label="Search
courses" placeholder="■  Search courses..."></div>
<section class="hero"><div><small>YOUR NEXT CHAPTER STARTS HERE</small><h1>Learn skills. Build your
future.</h1><p>Discover practical courses designed to help you grow at your own pace.</p></div><div
class="hero-badge">■</div></section>
<section id="progress"><h2>Your learning overview</h2><div class="stats"><div
class="stat"><small>Courses in progress</small><div class="number">03</div><div class="progress"><span
style="width:65%"></span></div></div><div class="stat"><small>Lessons completed</small><div
class="number">18</div><div class="progress"><span style="width:72%"></span></div></div><div
class="stat"><small>Learning streak</small><div class="number">7 days ■</div><div class="progress"><span
style="width:48%"></span></div></div></div></section>
<section id="courses"><h2>Popular courses</h2><div class="courses">
<article class="course"><div class="thumb">■</div><div class="category">WEB DEVELOPMENT</div><h3>HTML &
CSS Foundations</h3><p>Build clean, responsive webpages from scratch.</p><div class="meta"><span>■■■
Maya Sharma</span><span>★ 4.9</span></div><div class="progress"><span style="width:40%"></span></
div><p>Course progress · 40%</p><a class="enroll" href="#instructors">View course</a></article>
<article class="course"><div class="thumb green">■</div><div class="category">PROGRAMMING</
div><h3>Python for Beginners</h3><p>Learn variables, conditions, loops, and functions.</p><div
class="meta"><span>■■■ Rahul Verma</span><span>★ 4.8</span></div><div class="progress"><span
style="width:15%"></span></div><p>Course progress · 15%</p><a class="enroll" href="#instructors">View
course</a></article>
<article class="course"><div class="thumb orange">■</div><div class="category">AI & DATA</div><h3>AI
Essentials</h3><p>Explore AI concepts, data, and real-world use cases.</p><div class="meta"><span>■■■
Sara Khan</span><span>★ 4.7</span></div><div class="progress"><span style="width:0%"></span></
div><p>Course progress · Not started</p><a class="enroll" href="#instructors">Explore course</a></
article>
</div></section>
<section id="instructors"><h2>Meet your instructors</h2><div class="stats"><div class="stat"><b>Maya
Sharma</b><p style="color:#718096">Frontend developer · HTML & CSS</p></div><div class="stat"><b>Rahul
Verma</b><p style="color:#718096">Software engineer · Python</p></div><div class="stat"><b>Sara Khan</
b><p style="color:#718096">Data educator · AI fundamentals</p></div></div></section>
<p style="font-size:12px;color:#8490a5;margin-top:30px">Demo interface only. Course buttons are sample
navigation; actual video lessons, enrollment, and saved progress require additional functionality.</p>
</main></div></body></html>
```